MPH Public Health Data Accelerator

Data Analytics Starter Kit

Excel · SQL · Python

Installation guides and practice exercises for accelerator participants

Examples drawn from CDC Ebola Outbreak History Dataset

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DATASET USED IN THIS STARTER KIT

CDC Ebola Outbreak History — `ebola_outbreak_history_CDCDownload2026.xlsx`

53 Ebola outbreak records from 1976–2022 across two sheets: All Outbreaks (Year, Country/Region, Timing, Species, Cases, Deaths, Fatality Rate) and Summary by Country. Use the copy provided at the attached link or you may download from cdc.gov.

Useful Book References

1. **Excel 2021:** A Complete Step-by-Step Illustrative Guide from Beginner to Expert. Master the Essential Functions and Formulas in Less Than 20 Minutes a Day. Includes Tips & Tricks by Mark Thompson
2. **Learning SQL:** Generate, Manipulate, and Retrieve Data by Alan Beaulieu.
3. **Practice SQL problems:** SQL Practice Problems: 57 beginning, intermediate, and advanced challenges for you to solve using a “learn-by-doing” approach by Sylvia Moestl Vasilik
4. **Python for Data Analysis:** Data Wrangling with pandas, NumPy, and Jupyter by Wes McKinney.
   The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios by Steve Wexler, Jeffrey Shaffer, et al.

01 Microsoft Excel

Excel is the starting point for most public health data work. Check your applications first — it may already be installed.

How to Install

- Go to **microsoft365.com** and sign in with your school or work email. Many universities and employers provide Microsoft 365 free to employees / enrolled students.
- Free alternative: sheets.google.com — runs in any browser, no install needed.

Excel Exercise

Exercise 1: Filter and Count by Species

1. Open ebola_outbreak_all_outbreaks.xlsx.
2. Click any cell in your data → Data → Filter. Dropdown arrows appear on each column.
3. Click the dropdown on the Species column → uncheck all → check only Zaire ebolavirus → OK.
4. Observe: Excel now shows only Zaire ebolavirus rows.
5. In an empty cell (e.g., I2), type this formula and press Enter:

```
=COUNTIF(D:D,"Zaire ebolavirus")
```

6. Expected result: 35. Try it for: Sudan ebolavirus, Bundibugyo ebolavirus.

02 SQL with DBeaver

SQL is how you retrieve and analyze data in databases. We use DBeaver Community Edition, it's free, connects to nearly every database type, and the tool you're most likely to encounter in professional public health environments.

How to Install DBeaver

1. Go to dbeaver.io → click Download → select Community Edition (free). Choose your OS: Windows (.exe) or Mac (.dmg). Run the installer with all defaults.
2. Open DBeaver. Click Database → New Database Connection → select SQLite → Next. Under Path, click Create → save a file named ebola_practice.db → Finish.
3. Save the Ebola dataset as CSV: open the .xlsx in Excel → File → Save As → CSV (Comma delimited) → save as ebola_outbreaks.csv.
4. If necessary, in DBeaver: right-click your database → Connect.
5. In DBeaver: right-click your database → Import Data → CSV → Next → browse to ebola_outbreaks.csv → Next → set table name to ebola_outbreaks → Finish.
6. Open the SQL editor: right-click your database → SQL Editor → Open SQL Script. Run queries with Ctrl+Enter (Windows) or Cmd+Enter (Mac).

SQL Exercise

Exercise 1: SELECT and Filter with WHERE

Run each query in DBeaver. Press Ctrl+Enter (Cmd+Enter on Mac) to execute.

Query 1 — Preview your data:

```
SELECT * FROM ebola_outbreaks LIMIT 10;
```

Query 2 — Show only large outbreaks (over 100 cases), sorted largest first:

```
SELECT Year, Country_Region, Cases, Deaths
FROM ebola_outbreaks
WHERE Cases > 100
ORDER BY Cases DESC;
```

Question: Which outbreak had the most cases? What year did it occur?

03 Python

Python is a programming language used through Jupyter Notebooks to write and run code one block at a time and see results immediately. Install Anaconda to get Python, Jupyter, and all data libraries in one download. Another option is to use Google Colab.

How to Install Anaconda

1. Go to anaconda.com/download → download Anaconda Individual Edition for your OS (~600MB).
2. Run the installer. Accept all defaults. Do not change the installation path.
3. Open Anaconda Navigator → click Launch under Jupyter Notebook. Your browser opens to the Jupyter file browser.
4. Navigate to the folder with your Ebola dataset. Click New → Python 3 (ipykernel). A blank notebook opens.
5. Confirm setup: type the code below in the first cell and press Shift+Enter. If you see Python is ready! you're good to go!

```
import pandas as pd import
matplotlib.pyplot as plt
print('Python is ready!')
```

Python Exercise

Exercise 1: Load and Explore the Ebola Dataset

Run each block in a separate Jupyter cell (Shift+Enter to run).

Block 1 — Load the data:

```
df = pd.read_excel('ebola_outbreak_history_CDCDownload2026.xlsx',
sheet_name='All Outbreaks') df.head()   # view first 5 rows
df.shape    # should return (53, 7)
```

Block 2 — Summary statistics:

```
df[['Cases', 'Deaths', 'Fatality Rate (%)']].describe()
```

Block 3 — Count outbreaks by species:

```
df['Species'].value_counts()
```

Question: How many unique Ebola species appear in the dataset?

Note: If you get an ImportError, run !pip install openpyxl in a new cell first.

CONGRATULATIONS, YOU'VE COMPLETED THE INTRODUCTION TO THE ACCELERATOR PROGRAM!!

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